

ECOTEC_Online Student Risk Dataset

Preprocessing and Reproducibility Description

This document describes the data acquisition, target construction, validation, cleaning, anonymization, preprocessing, and benchmark-reproducibility workflows associated with the ECOTEC_Online Student Risk Dataset.

1. Data Acquisition

Institutional records were extracted from academic management systems associated with online higher education programs at Universidad ECOTEC, Ecuador. The dataset integrates demographic, academic, financial, admission, and institutional variables. The released data represent a cross-sectional institutional snapshot.

2. TARGET_RISK Construction and Temporal Interpretation

TARGET_RISK is a binary administrative target derived during institutional curation from the restricted institutional source variables STUDENT_TYPE and GRADUATION_DATE. Class 0 indicates that alumni status or a valid graduation date was recorded; class 1 indicates that the student was not recorded as alumni and no graduation date was available.

The target was created from administrative information available in the same institutional snapshot as the released predictors. It is therefore contemporaneous and does not constitute a prospective dropout or retention outcome. No future observation window is encoded in the target.

STUDENT_TYPE and GRADUATION_DATE were excluded from the public release and from the benchmark predictor matrix after target construction. Consequently, the public benchmark inputs do not directly include the two restricted fields used to define the target.

3. Initial Validation

The preprocessing workflow includes automated validation of required variables, target labels, dataset dimensions, and file integrity. Validation scripts verify the presence of expected columns and the consistency of the binary target labels.

4. Data Cleaning

Cleaning procedures include duplicate checks, normalization of variable names, validation of target completeness, and formatting harmonization for reproducible analytical workflows. The released dataset contains complete observations for all published variables.

5. Data Anonymization

Personally identifiable information was removed before public dissemination. The anonymization process excluded names, identification numbers, email addresses, telephone numbers, and direct institutional identifiers. Additional transformations and categorical aggregations were applied to reduce re-identification risk while preserving analytical utility.

6. Feature Engineering and Encoding

Benchmark scripts implement preprocessing pipelines using scikit-learn transformers. Categorical variables are encoded using OneHotEncoder. Numerical imputation and categorical imputation remain available as defensive pipeline safeguards; because the released dataset contains no missing values, these steps do not replace any observed values in the archived release. Scaling is applied where required by the benchmark model.

7. Benchmark Reproducibility Pipelines

The repository includes reproducible Random Forest and Logistic Regression benchmark workflows implemented in Python. The scripts use a stratified 70%/30% train-test split and a fixed random seed of 42. These models are computational reproducibility baselines only. Their results should not be interpreted as prospective dropout or retention prediction, validated early-warning performance, institutional readiness, or operational decision support.

8. Software Environment

The preprocessing and benchmarking workflows were implemented using Python 3.11. Package dependencies and versions are documented in the repository requirements.txt file.

9. FAIR and Open Science Considerations

The repository was organized according to FAIR data principles and includes metadata documentation, preprocessing scripts, benchmark resources, outputs, and citation files intended to support open-science dissemination.

10. Repository Components

- Curated anonymized public dataset
- Processed reproducibility dataset
- Preprocessing summary
- Variable dictionary and codebook
- Preprocessing documentation
- Benchmark scripts and outputs
- Figure-generation scripts
- Requirements file
- FAIR and citation metadata

Citation

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